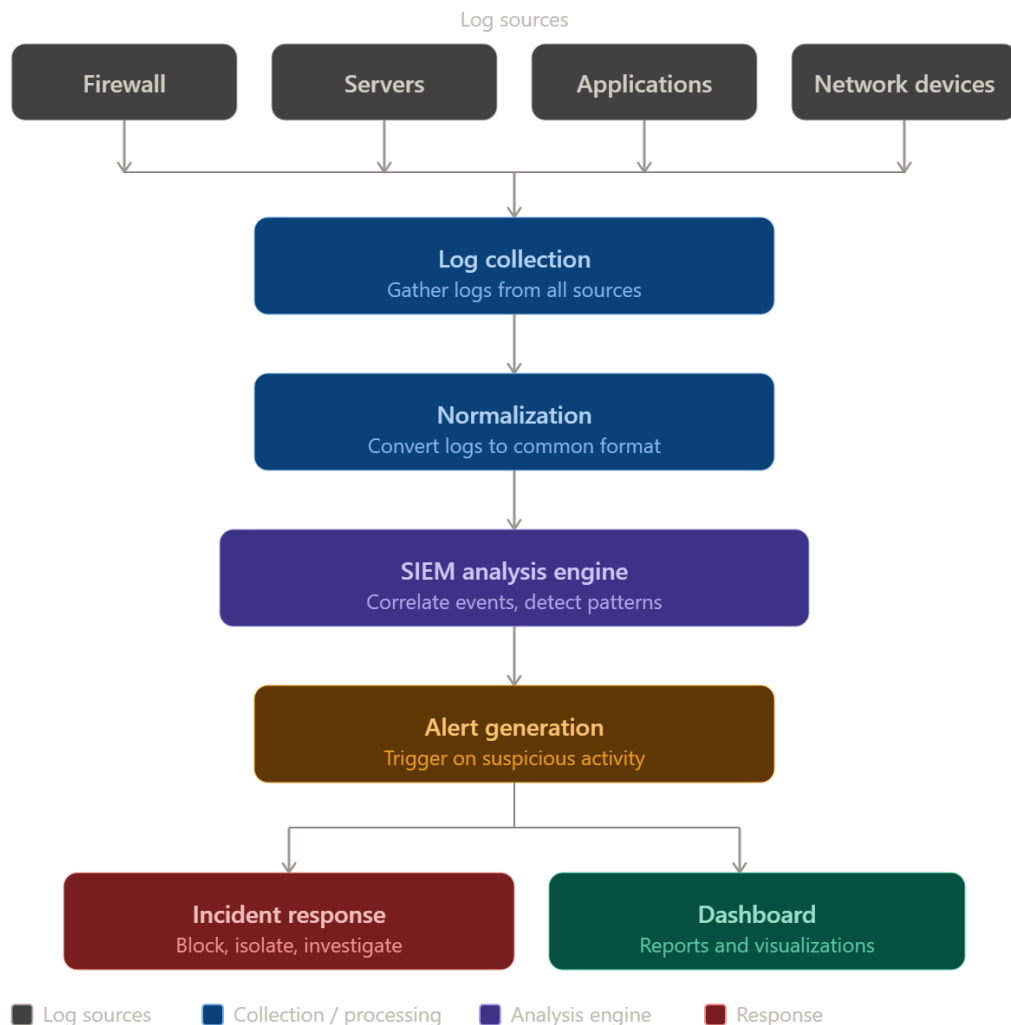


Day 6- SIEM

Draw a SIEM workflow diagram showing log collection, analysis, and alerting.



Task

1. The workflow works in the following stages:

First, logs are collected from all devices on the network such as firewalls, servers, routers, and applications. Every activity that happens on these devices is recorded as a log entry.

Second, all these logs are normalized which means they are converted into a common format so the system can read and compare them easily even if they came from different types of devices.

Third, the SIEM analysis engine processes all the normalized logs and looks for patterns, correlations, and suspicious behavior by comparing current activity against predefined rules.

Fourth, when something suspicious is detected the system generates an alert and notifies the security team immediately so they can take action.

Finally, the security team performs incident response by investigating the alert, blocking the attacker, isolating affected systems, and documenting the incident. All findings are also displayed on a dashboard for monitoring and reporting.

2. Three Popular SIEM Tools

1. **Splunk** is one of the most widely used SIEM tools in the world. It collects and indexes large amounts of log data and provides powerful search and visualization features. It is used by large organizations and government agencies for real time security monitoring.

2. **IBM QRadar** is an enterprise level SIEM tool that automatically collects logs, detects threats, and prioritizes alerts based on their severity. It is known for its ability to handle very large networks and is widely used in banks and financial institutions.

3. **ELK Stack (Elasticsearch, Logstash, Kibana)** is a very popular open source SIEM solution made up of three tools that work together:

Elasticsearch is the search and storage engine that stores all the collected log data and allows very fast searching through millions of log entries within seconds.

Logstash is the data processing tool that collects logs from different sources, filters them, and sends them to Elasticsearch in a normalized format. It acts as the pipeline between the log sources and the storage engine.

Kibana is the visualization tool that sits on top of Elasticsearch and provides a visual dashboard where security analysts can see charts, graphs, and real time data about what is happening on the network.

3. Real World Detection Example

Scenario: Brute Force Attack Detected by SIEM

In a real organization a SIEM tool like Splunk was monitoring the authentication logs of all servers. The SIEM had a rule configured that said if more than 5 failed login attempts happen from the same IP address within 1 minute an alert should be triggered.

One day the SIEM detected that the IP address 192.168.1.10 had made 47 failed login attempts for the root account within just 60 seconds. The SIEM immediately generated a high severity alert and notified the security team.

The security team investigated the alert on the dashboard, confirmed it was a brute force attack, and blocked the IP address using the firewall. They also reset the affected account password and added two factor authentication to prevent future attacks.

This is a perfect example of how SIEM tools help organizations detect and stop attacks in real time before they cause serious damage.